MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 11** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. From the top of a 20 m high building, the angle of elevation of the top of a tower is $60°$ and the angle of depression of its base is $30°$. The horizontal distance between the building and the tower is: [1]

(A) $20\sqrt{3}$ m (B) $\dfrac{20}{\sqrt{3}}$ m (C) 40 m (D) 60 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A)
Justification: From 20 m building top, base depression $= 30° \implies$ horizontal distance $d = 20\cot 30° = \boxed{20\sqrt{3}\text{ m}}$.

Q2. In Q1, the total vertical height of the tower is: [1]

(A) 60 m (B) 80 m (C) $20\sqrt{3}$ m (D) 100 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B)
Justification: From Q1, $d = 20\sqrt{3}$ m. Tower segment above building $= d\tan 60° = (20\sqrt{3})\sqrt{3} = 60$ m. Total tower height $= 20 + 60 = \boxed{80\text{ m}}$.

Q3. An electric pole is 10 m high. A steel wire tied to the top of the pole is affixed to the ground at an angle of $45°$. The length of the wire is: [1]

(A) $10\sqrt{2}$ m (B) $10\sqrt{3}$ m (C) 20 m (D) $5\sqrt{2}$ m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A)
Justification: $\sin 45° = \dfrac{10}{L} \implies L = \boxed{10\sqrt{2}\text{ m}}$.

Q4. A steel wire of length 20 m is tied from the top of an electric pole to the ground making an angle of $60°$. The height of the pole is: [1]

(A) $10\sqrt{3}$ m (B) 10 m (C) $20\sqrt{3}$ m (D) $\dfrac{20}{\sqrt{3}}$ m

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $h = 20\sin 60° = 20 \times \dfrac{\sqrt{3}}{2} = \boxed{10\sqrt{3} \text{ m}}$.

Q5. **Assertion (A):** As an observer walks towards the foot of a vertical monument, the angle of elevation of the top of the monument continuously decreases.

Reason (R): As distance d decreases, $\tan\theta = \dfrac{h}{d}$ increases, which implies θ increases for acute angles. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D)

Justification: Walking towards the monument decreases d, which makes $\tan\theta = \dfrac{h}{d}$ increase and hence makes θ increase (not decrease). Thus (A) is false and (R) is true.

SECTION B: Very Short Answer Questions

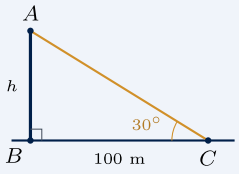
MATHS MANTHAN | ($2 \times 2 = 4$ Marks)

Q6. If the angle of elevation of the top of a tower from a distance of 100 m from its foot is $30°$, find the height of the tower. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let tower $AB = h$ m, distance $BC = 100$ m, $\angle ACB = 30°$.

[0.5 Mark]



$$\tan 30° = \frac{h}{100} \implies h = \frac{100}{\sqrt{3}}$$

$$\implies h = \frac{100\sqrt{3}}{3} = \boxed{\frac{100\sqrt{3}}{3} \text{ m} \approx 57.73 \text{ m}}$$

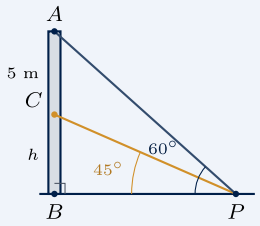
Computing tower height: [1.5 Marks]

Q7. A vertical flagstaff of height 5 m is fixed on top of a tower. From a point on the ground, the angles of elevation of the top and bottom of the flagstaff are 60° and 45° respectively. Find the height of the tower. [Take $\sqrt{3} = 1.732$] **[2]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Let tower be $BC = h$ m and flagstaff be $AC = 5$ m, observed from P with $\angle CPB = 45°$ and $\angle APB = 60°$. [0.5 Mark]

In $\triangle CBP$: $BP = h$; in $\triangle ABP$: $\sqrt{3} = \dfrac{h+5}{h} \implies h = \dfrac{5}{\sqrt{3}-1} = 2.5(\sqrt{3}+1) = \boxed{6.83 \text{ m}}$. [1.5 Marks]



SECTION C: Short Answer Questions

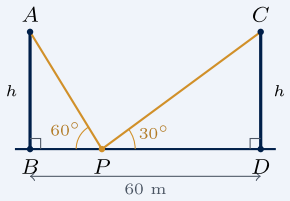
MATHS MANTHAN | $(2 \times 3 = 6$ Marks)

Q8. Two lamp posts of equal heights are situated on opposite sides of a road 60 m wide. From a point on the road between them, the angles of elevation of the tops of the posts are 60° and 30°. Find the height of the posts and the position of the observation point. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let equal posts $AB = CD = h$ m stand across 60 m road BD, and point P satisfy $BP = x$ m, $PD = (60 - x)$ m with angles $60°, 30°$. [0.5 Mark]

In $\triangle ABP$: $h = \sqrt{3}\,x$; in $\triangle CDP$: $h = \dfrac{60-x}{\sqrt{3}}$. [1 Mark]



Equating: $3x = 60 - x \implies 4x = 60 \implies x = 15$ m, $PD = 45$ m. Then $h = 15\sqrt{3} = 15(1.732) = 25.98$ m. [1 Mark]

$$\therefore \text{ Height} = \boxed{15\sqrt{3} \text{ m} \approx 25.98 \text{ m}}$$

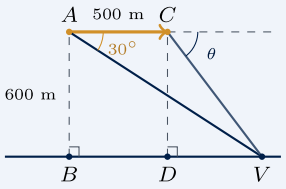
$$\text{Position} = \boxed{15 \text{ m from post } AB \text{ (and 45 m from } CD)}$$

[0.5 Mark]

Q9. A surveillance helicopter hovers stationary at an altitude of 600 m. The angle of depression of a stationary patrol vessel is 30°. If the helicopter flies horizontally towards the vessel at a speed of 180 km/h, find the angle of depression after 10 seconds. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let helicopter be initially at A (above B, height 600 m) observing vessel V at depression $30°$, so $BV = 600\cot 30° = 600\sqrt{3}$ m. [0.5 Mark]



Speed $= 180$ km/h $= 180 \times \dfrac{5}{18} = 50$ m/s. In 10 s, horizontal distance flown towards V is $AC = BD = 50 \times 10 = 500$ m. [1 Mark]

Remaining horizontal distance from D (beneath new position C) to vessel V:

$$DV = 600\sqrt{3} - 500 \approx 600(1.732) - 500 = 1039.2 - 500 = 539.2 \text{ m}$$

[0.5 Mark]

Let new angle of depression be θ: $\tan\theta = \dfrac{600}{600\sqrt{3}-500} = \dfrac{6}{6\sqrt{3}-5} \approx 1.1128$. Thus exact angle is:

$$\theta = \boxed{\tan^{-1}\left(\frac{6}{6\sqrt{3}-5}\right) \approx 48.05°}$$

[1 Mark]

SECTION D: Long Answer / HOTS Questions — MATHS MANTHAN | $(2 \times 5 = 10$ Marks)

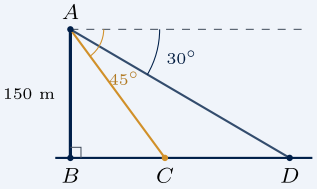
Q10. An observer on top of a cliff 150 m high observes two boats in the sea moving directly towards the foot of the cliff along the same radial line. If the angles of depression of the boats change from $30°$ to $45°$ and $45°$ to $60°$ respectively, find the distances travelled by both boats. [Take $\sqrt{3} = 1.732$] **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let cliff $AB = 150$ m and ground distances from foot B at angles $30°, 45°, 60°$ be:

$$x_{30} = 150\sqrt{3} \text{ m}, \qquad x_{45} = 150 \text{ m}, \qquad x_{60} = \frac{150}{\sqrt{3}} = 50\sqrt{3} \text{ m}$$

[2 Marks]



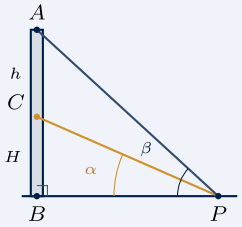
Distance travelled by Boat 1 ($30° \to 45°$): $d_1 = x_{30} - x_{45} = 150\sqrt{3} - 150 = 150(1.732 - 1) = \boxed{109.8 \text{ m}}$. [1.5 Marks]

Distance travelled by Boat 2 ($45° \to 60°$): $d_2 = x_{45} - x_{60} = 150 - 50\sqrt{3} = 150 - 50(1.732) = 150 - 86.6 = \boxed{63.4 \text{ m}}$. [1.5 Marks]

Q11. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height h. At a point on the plane, the angles of elevation of the bottom and top of the flagstaff are α and β respectively. Prove that the height of the tower is: $\dfrac{h\tan\alpha}{\tan\beta-\tan\alpha}$. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let tower be $BC = H$ surmounted by flagstaff $AC = h$, observed from point P on the horizontal plane at distance $BP = x$, with $\angle CPB = \alpha$ and $\angle APB = \beta$. [1 Mark]



In right $\triangle CBP$: $\tan\alpha = \dfrac{H}{x} \implies x = \dfrac{H}{\tan\alpha}$ — (i). In right $\triangle ABP$: $\tan\beta = \dfrac{H+h}{x} \implies x = \dfrac{H+h}{\tan\beta}$ — (ii). [2 Marks]

Equating (i) and (ii): $\dfrac{H}{\tan\alpha} = \dfrac{H+h}{\tan\beta} \implies H\tan\beta = H\tan\alpha + h\tan\alpha.$

$$H(\tan\beta-\tan\alpha) = h\tan\alpha \implies \boxed{H = \frac{h\tan\alpha}{\tan\beta-\tan\alpha}} \quad \text{[Hence Proved]}$$

Algebraic isolation of H: [2 Marks]

SECTION E: Applied Case Study

MATHS MANTHAN | $(1 \times 5 = 5$ Marks$)$

Q12. CASE STUDY: Cyclonic Storm and Coconut Palm Fracture

A vertical coconut palm tree snapped at height B without detaching from trunk base A. Its crown top C struck flat ground 12 m horizontally from root A, making an angle of $30°$ with the ground.

(i) Find the vertical height above the ground at which the palm tree snapped (AB). [Take $\sqrt{3} = 1.732$] [1]

(ii) Find the length of the broken trunk section that bent over to touch the ground (BC). [2]

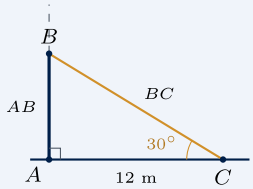
(iii) Calculate the total original standing height of the coconut palm tree before the cyclonic storm. [2]

OR

If the tree had broken such that its top struck the ground at an angle of $45°$ at a distance of 8 m from its root, find its original standing height. [Take $\sqrt{2} = 1.414$] [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

(i) In right $\triangle BAC$ ($\angle BCA = 30°$, $AC = 12$ m): [1 Mark]



$$\tan 30° = \frac{AB}{12} \implies AB = \frac{12}{\sqrt{3}} = 4\sqrt{3} = 4(1.732) = \boxed{6.928 \text{ m}}$$

(ii) Bent trunk section length BC: [2 Marks]

$$\cos 30° = \frac{12}{BC} \implies BC = \frac{24}{\sqrt{3}} = 8\sqrt{3} = 8(1.732) = \boxed{13.856 \text{ m}}$$

(iii) Main Choice: Total original standing height $= AB + BC = 4\sqrt{3} + 8\sqrt{3} = 12\sqrt{3} = 12(1.732) = \boxed{20.784 \text{ m}}$. [2 Marks]

(iii) OR Choice: At $45°$ and 8 m from root: $AB' = 8\tan 45° = 8$ m, $B'C' = \dfrac{8}{\cos 45°} = 8\sqrt{2}$ m. Original height $= 8(1+\sqrt{2}) = 8(1+1.414) = \boxed{19.312 \text{ m}}$. [2 Marks]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★